best practices.[56] Furthermore, Building Block 19 reiterates the significance of adaptive governance and suggests the monitoring and review of the implementation of the framework so as to ensure its compliance with contemporary challenges.[57]

Overall, the Building Blocks revolve thematically around the concepts of appropriation and sharing of benefits and advocate in favor of a level-playing field among the stakeholders involved in space resource activities.

III. CHALLENGES WITH REGARD TO THE GOVERNANCE OF SPACE RESOURCE ACTIVITIES

The questions surrounding the governance of space resources formed the fundamental element in establishing The Hague Working Group in 2016 and has become part of international discussions in the recent years. In 2017 the UNCOPUOS introduced in the agenda of the Legal Subcommittee an item titled "General Exchange of Views on Potential Legal Models for Activities in Exploration, Exploitation and Utilization of Space Resources."[58] And in the meantime two countries, the United States and Luxembourg, have adopted national legislation dedicated to the regulation of space resource activities.[59]

The interest in the governance of space resources was sparked by the emerging commercial initiatives, but its practical dimensions were not understood from the beginning. The legality of the use of space resources for profit was initially contested on the grounds of the non-appropriation principle, which prohibits the occupation, sovereignty and any other means of appropriation in outer space. Furthermore, the prospects for excessive proceeds from the commercial utilization of space resources was seen as contradictory to the duty of States to share the benefits from the conduct of space activities. The goal of the Working Group in this regard

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[56] BUILDING BLOCKS, *supra* note 7, ¶ 17.

[57] *Id.,* ¶19.

[58] Comm. on the Peaceful Uses of Outer Space, Rep. of the Legal Subcomm. on Its Fifty-Fifth Session, ¶ 250, U.N. Doc. A/AC.105/1113 (2016).

[59] *See* U.S. Commercial Space Launch Competitiveness Act, Pub. L. No. 114-90, § 401, 129 Stat. 70 (2015); Loi 674 du 20 juillet 2017 sur l'exploration et l'utilisation des resources de l'espace [Law 674 of July 20, 2017 on the Exploration and Use of Space Resources], JOURNAL OFFICIEL DU GRAND DUCHE DE LUX,, July 28, 2017, *available at* http://legilux.public.lu/eli/eta1/leg/loi/2017/07/20/a674/jo.

was to create a multi-stakeholder platform of discussion, which would not be limited to the legal and policy aspects of space resource utilization, but would also encompass the corresponding technical and business perspectives. This inclusive approach contributed to the formulation of Building Blocks in a manner which combines accordance with international legal obligations with the technical, scientific and commercial facets of space resource activities.

The governance of space resource activities was foreseen by the Moon Agreement,[60] which called for the establishment of a regime to govern the exploitation of mineral resources. However, due to the lack of wide acceptance of the Moon Agreement, this regime was not seen as prerequisite for the undertaking of such activities on national and international level. Regardless of any treaty indication, the necessity for a governance scheme derives from the need for legal certainty and for compliance with the existing international legal framework. The sustainable development of space resource activities depends on the foreseeability of the legal obligations borne for the operators, but the general language of the space law principles requires further interpretation of the terms at hand. In favor of striking balance among the interests of the various stakeholders, a collaborative approach for the common perception of the treaty terms is essential. As far as the Working Group is concerned, the interdisciplinary analysis of ambiguous terms, such as sharing of benefits and rights to resources is a way to achieve common grounds. Even though the purpose of the Building Blocks is not to propose specific obligations for States or private parties, they aim to provide recommendations with regard to the implementation of the treaty provisions and the approach of the States and intergovernmental organizations to the governance of space resource activities.

As these lines are written, the discussion on the implementation strategy of the Building Blocks is currently ongoing among the participants of the Working Group. Different views have been expressed in relation to the regulatory or non-legally binding character of the governance framework, as well as its negotiation procedures. Without prejudice to the outcome of these discussions, the

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[60] *See* Moon Agreement, *supra* note 13, art. 11.

Working Group acknowledges the importance of multilateral approach and international dialogue, as well as of any State and other initiative to govern space resource activities.

IV. THE WAY FORWARD

During its four-year operation, the Working Group has received insightful feedback with regard to the various aspects of space resource activities. The extensive volume of the comments received from members, observers, the panels and external parties have been taken into consideration for the improvement of the text of the Building Blocks. In order to preserve the input that was not reflected in their text, as well as to provide background information about their formulation and language selection, the Working Group plans to compile a Commentary of the Building Blocks.[61]

The purpose of the Working Group will be fulfilled to the extent that the Building Blocks serve as recommendations for a framework for the governance of space resource activities. Owing to the diversity of background and professional expertise of its participants, the Working Group has opted to develop a concise set of guidelines, which combine the legal implications of space resource utilization, the interests of industrial stakeholders and the advancement in science and technology. This multi-spectrum approach has allowed for an inclusive discussion on the challenges raised in this new field of space activities.

The Hague International Space Resources Governance Working Group is scheduled to conclude its activities at the end of 2019. It is currently working toward finalizing its main deliverables, namely the Building Blocks and the recommendations regarding their implementation. The final version of the Building Blocks is not expected to divert significantly from the published draft version. Even though the Working Group has not yet indicated the appropriate fora for further discussion on the Building Blocks, it anticipates to incentivize government initiatives for the negotiation of a framework on the governance on space resource activities on multilateral or national level. In this end, it has organized and will continue planning outreach activities and encourage its participants to

[61] The details regarding the publication of the Commentary will be posted on the website of the Working Group when they become available.

promote the work of the Group in their respective network. The negotiation strategies and the format of the potential framework depend on the stakeholders, be it States, intergovernmental organizations or other institutions, that will be interested in introducing the topic of space resources governance through official channels. In the meantime, the Working Group will communicate its developments via its website, social media and newsletter.

INTERNATIONAL MANAGEMENT OF SPACE RESOURCE EXTRACTION: DON'T PUT THE CART BEFORE THE HORSE

Michael Weinhoffer[*]

ABSTRACT

Commercial space mining is forecasted to be an extremely lucrative business, given the presence of high-value minerals on near-Earth asteroids and the potential to use such materials to further space exploration. Despite such potential, no commercial space mining missions, or even scouting missions to the Moon or near-Earth asteroids, have taken place at publication. Even worse, two of the primary commercial space mining companies in the world, Planetary Resources and Deep Space Industries, were both bought out in late 2018, and all company activities appear to be on hold. Because of the lack of missions and technology development struggles, it would be inappropriate to implement a legally binding space mining agreement at this time. Doing so would stifle innovation and may even put the final "nail in the coffin" for the industry. Furthermore, any space mining agreement should incorporate elements from deep seabed mining and telecommunications law to minimize overexploitation of space resources.

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I. INTRODUCTION

Both asteroid and lunar mining involve the extraction of water or minerals from either the asteroid or the Moon's surface, as the case may be. The majority of asteroids in the solar system can be classified as C, S, or M type. The United Kingdom's Asteroid Mining Corporation notes that C-type asteroids are mainly composed of carbon and other common minerals, but also may contain sizable amounts of extractable water ice. S-type asteroids are composed of silicon, iron and magnesium minerals, while M-type asteroids are composed of nickel, iron, gold and platinum-group metals such as platinum and iridium.[1] Additionally, India's Chandrayaan-1 and the US National Aeronautics and Space Administration's (NASA) LCROSS spacecraft both independently discovered liquid water ice on the Moon's surface, making it another attractive space mining location.[2]

The three major United States (US) companies that are actively developing space resource extraction technologies are Planetary Resources, Inc.[3] (Planetary Resources), Moon Express, Inc.[4] (Moon Express) and the Shackleton Energy Company, Inc.[5] (Shackleton Energy Company). All three companies are headquartered in the US and are less than fifteen years old. The space resource industry is more nascent than the commercial launch industry, however, because of the economic lure of mining the Moon and asteroids, more space resource extraction companies will likely form in the coming years. Harvesting rare minerals from asteroids or the Moon and then using them to help refuel spacecraft or build outer space infrastructure could provide an investment return of billions

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[1] *Why Mine Asteroids?* ASTEROID MINING CORP., http://asteroidminingcorporation.co.uk/why-mine-asteroids (last visited May 7, 2019).

[2] *See* Anthony Colaprete, et al., *Detection of Water in the LCROSS Ejecta Plume. Science*, SCIENCEMAG.ORG (Oct. 22, 2010), https://pdfs.semanticscholar.org/ba89/d994f2b1c65c5b64662ca24d04d257f40936.pdf?_ga=2.7275387.838416552.1566355003-1729732047.1566355003; C.M. Pieters, et al., *Characterization of Spatial Distribution of OH/H_2O on the Surface of the Moon Seen by M^3 on Chandrayaan-1*. SCIENCE EXPRESS (Sep. 24, 2009), http://repository.ias.ac.in/14506/1/307.pdf.

[3] PLANETARY RESOURCES, https://www.planetaryresources.com/ (last visited June 12, 2018).

[4] MOON EXPRESS, http://www.moonexpress.com/ (last visited June 14, 2018).

[5] SHACKLETON ENERGY CO., http://www.shackletonenergy.com/ (last visited June 14, 2018).

of dollars. Of course, the true profitability of space resource extraction cannot be determined until an extraction mission occurs – something that is not likely to happen before the end of this decade.

According to Planetary Resources, there more than 16,000 asteroids that could be ideal candidates for resource harvesting due to their proximity to Earth.[6] Furthermore, the company estimates that there are collectively two trillion tons of water available on those asteroids.[7] The company has built and operated two technology demonstration missions, Arkyd-3R and Arkyd-6. Arkyd-3R, a three-unit CubeSat, was deployed from the International Space Station on July 15, 2015.[8] A follow-on six-unit CubeSat, Arkyd-6, was launched on an Indian rocket on January 11, 2018.[9] This satellite featured the first commercial mid-wave infrared imager to be operated in space, which will be used on later missions to search for water within asteroids.[10] A third mission, Arkyd-301, is currently under development. This mission involves launching multiple "asteroid prospecting" satellites on a single rocket. Each satellite will analyze a certain number of near-Earth asteroids to determine whether there is accessible water.[11] Planetary Resources then plans to extract water for the creation of rocket propellant or for radiation shielding, along with other Earth-focused purposes. Unfortunately, Planetary Resources is having difficulty attracting enough investors to fund their commercial asteroid scouting and mining plans. The company announced on October 31, 2018, that it was acquired by "blockchain venture production studio" ConsenSys, Inc.[12] Although Planetary Resources' role in the new company is uncertain,

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[6] *Why Asteroids*, PLANETARY RESOURCES, https://www.planetaryresources.com/why-asteroids/ (last visited Aug. 7, 2019).

[7] *Id.*

[8] Mike Wall, *Planetary Resources' Asteroid-Mining Goals Move Closer with Satellite Launch,* SPACE.COM (Jan. 12, 2018), https://www.space.com/39363-planetary-resources-asteroid-mining-satellite-launches.html.

[9] *Id.*

[10] *Mission Success: Arkyd-6 Tests Key Technologies For Commercial Space Resource Exploration*, PLANETARY RESOURCES (Apr. 24, 2018), https://www.planetaryresources.com/2018/04/mission-success-arkyd-6-tests-key-technologies-for-commercial-space-resource-exploration/.

[11] *About the Exploration Program*, PLANETARY RESOURCES, https://www.planetaryresources.com/missions/arkyd-301/ (last visited Aug. 22, 2019).

[12] Alan Boyle, *Why in the Universe is a Blockchain Company Buying the Assets of a Formerly High-Flying Asteroid Miner?* GEEKWIRE (Oct. 31, 2018),

they were the first deep-space mining company to launch their own spacecraft into orbit and are the only ones to do so at present.

Moon Express is developing missions to extract resources from the Moon. According to company material, "the Moon is Earth's 8th continent, a new frontier for humanity with precious resources that can bring enormous benefits to life on Earth and our future in space."[13] The company plans to launch their first lunar lander to the Moon in July 2020.[14] Moon Express was a participant in the Google Lunar XPRIZE competition, which was a challenge for a private company to soft-land a lander on the surface of the Moon, deploy and drive a rover 500 meters, and transmit data by March 31, 2018.[15] The first private company to complete the challenge would have received a $20 million prize, but unfortunately no company, including Moon Express, was able to launch their spacecraft by the competition deadline, so the prize went unclaimed.[16]

Shackleton Energy Company also plans to harvest water from the Moon to create rocket fuel for use in rockets around Earth's orbit and to ultimately power them to deep space.[17] The company has an extensive plan to sell rocket fuel partly made from the Moon's water to customers in Low Earth Orbit, but no descriptions of the harvesting vehicles have been released.[18] A large amount of lunar infrastructure is necessary to fulfill the company's mission, and flights to the Moon by Shackleton may be several years away.

There was a fourth US-based company that sought to developing technology for a variety of deep space activities, including space

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https://www.geekwire.com/2018/consensys-blockchain-studio-acquires-planetary-resources-asteroid-mining-venture/.

[13] *Why the Moon?* MOON EXPRESS, http://www.moonexpress.com/ (last visited Aug. 7, 2019).

[14] Jeff Foust, *Moon Express Raises $12.5 million*, SPACENEWS (Oct. 1, 2018), https://spacenews.com/moon-express-raises-12-5-million/.

[15] *How to Win,* XPRIZE FOUND., https://www.xprize.org/prizes/google-lunar (last visited Aug. 22, 2019).

[16] Loren Grush, *No One Won the Google Lunar X Prize, But These Competitors are Still Shooting for the Moon*, THE VERGE (Mar. 31, 2018), https://www.theverge.com/2018/3/31/17176530/google-lunar-x-prize-competition-spaceil-moon-express-astrobotic.

[17] *Overview,* SHACKLETON ENERGY CO., http://www.shackletonenergy.com/overview/ (last visited Aug. 7, 2019).

[18] *Program*, SHACKLETON ENERGY CO., http://www.shackletonenergy.com/program/#program1 (last visited Aug. 7, 2019).

mining. Founded in 2013, Deep Space Industries (DSI) was developing low-cost space exploration spacecraft and propulsion systems for small satellites. Like Planetary Resources, DSI was recently acquired by another company, Bradford Space, Inc., which develops space systems in Sweden and the Netherlands.[19] Unlike Planetary Resources, DSI was rebranded as Bradford Space, Inc. (BSI), and while it will continue to work on a satellite bus "intended for use on missions beyond Earth orbit," it will not pursue asteroid mining missions in the near future.[20]

On the international side, there are two notable commercial space mining ventures: iSpace and the Asteroid Mining Corporation. The Japanese company iSpace "aims to be a vehicle for companies on Earth to access new business opportunities on the Moon and ultimately incorporate the Moon into Earth's economic and living sphere."[21] The company managed Team HAKUTO, one of the five finalists in the Google Lunar XPRIZE competition, but the team was unable to complete the challenge.[22] The competition team's efforts involved into a full lunar exploration campaign, with the first robotic mission consisting of a soft landing on the Moon in 2021, followed by a second lander with a small rover in 2023.[23] Ultimately, the company plans to "deploy swarms of rovers to the lunar surface to pioneer the discovery and development of lunar resources . . . ."[24]

The Asteroid Mining Corporation, based in England, is taking a three-step approach to mining asteroid resources. In 2020, an Earth-orbiting satellite will scan Near-Earth Objects and determine the best candidates for further study. In 2023, a spacecraft will be sent to the most attractive asteroid to study its metallurgical properties. Finally, in 2028, a spacecraft will land on the asteroid, recover several tons of resources, and return them to Earth orbit for

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[19] Jeff Foust, *Deep Space Industries Acquired by Bradford Space,* SPACENEWS (Jan. 2, 2019), https://spacenews.com/deep-space-industries-acquired-by-bradford-space/.

[20] *Id.*

[21] *Project*, ISPACE, https://ispace-inc.com/project/ (last visited Aug. 25, 2019).

[22] *Id.*

[23] *Id.*

[24] *Id.*

later use, thus beginning the company's asteroid mining operations.[25]

Over the past few years, there has been an extensive amount of literature and discussion on the legality of commercial space mining activities[26]– there is significantly less dialog regarding the sustainable management of such activities. Even though no commercial space mining missions have been launched as of this writing, it is of great benefit to the international community to discuss specifically what can be done to manage both the harvestable resources themselves and the companies looking to extract them from asteroids or from the Moon.

This Note offers a strategy for the management of the space mining industry by the international community. It begins with an exploration of the legal tenets relevant to commercial space mining and recent related developments. Against this backdrop, it encourages the incorporation of successful practices from telecommunications, fisheries and deep-sea mining law into the governance system for private space mining. This proposal notwithstanding, this Note concludes by urging the international community to encourage space technology innovation by delaying the implementation of a binding international regulations until the industry reaches a maintainable level of maturity and stability.

II. LEGAL BACKGROUND OF SPACE RESOURCE EXTRACTION

A. Generally

All legal questions regarding both private and public outer space activities should first refer to the 1967 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (Outer

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[25] *Our Missions,* ASTEROID MINING CORP., https://asteroidminingcorporation.co.uk/our-missions (last visited Aug. 25, 2019).

[26] *See, e.g.,* Comm. on the Peaceful uses of Outer Space, Rep. of the Legal Subcomm. on Its Fifty-Eighth Session, Sec. XIII, U.N. Doc. A/AC.105/1203 (2019); *The Hague International Space Resources Governance Working* Group, LEIDEN UNIVERSITY, https://www.universiteitleiden.nl/en/law/institute-of-public-law/institute-for-air-space-law/the-hague-space-resources-governance-working-group (last visited Aug. 27, 2019; *Position Paper on Space Resource Mining,* INT'L INST. OF SPACE LAW (Dec. 20, 2015). http://www.iislweb.org/docs/SpaceResourceMining.pdf;) (see "Recent Developments").

Space Treaty or OST).[27] A number of international and space law experts have expressed the opinion "that the fundamental provisions of [the OST] are so well-observed and respected that they exist as an entirely different set of legal rules, outside of the textual treaty, as "customary" international law."[28] This means that even nations who have not ratified the OST adhere to those principles because of *opinio juris,* or a belief that they should.[29]

Russia, China and the US are the most notable parties to the OST, as they have invested the most substantially in space activities. The OST Articles that are most applicable to space mining are Articles I, II and VI. Article I mandates that "the exploration and use of outer space, including the Moon and other celestial bodies shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development," and that "there shall be free access to all areas of celestial bodies" by all nations.[30] Article II states that "[o]uter space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means."[31] Finally, Article VI of the treaty states that private space activities, such as commercial space mining operations, "shall require authorization and continuing supervision by the appropriate State Party to the Treaty."[32] These three Articles are among the most fundamental of the Outer Space Treaty, but proposed space mining activities have reignited international debate over their proper interpretation.

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[27] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space Including the Moon and Other Celestial Bodies art. II, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty].

[28] *See* Scott Hatton, *50 Years of the Outer Space Treaty*, INT'L INST. OF SPACE LAW, https://iislweb.org/50-years-of-the-outer-space-treaty/ (last visited Aug. 7, 2019).

[29] "Customary law requires an objective or material element – State practice (material facts) – and a subjective or psychological element – the belief that the practice is lawful (opinio juris)." Alexander Soucek, *Recalling Basics of Public International Law*, *in* IN SPACE LAW ESSENTIALS – VOLUME I 12 (2016) (internal quotations omitted). Soucek notes that the "three freedoms" granted by Article I of the Outer Space Treaty (i.e. freedom of access to, exploration and use of outer space) may have the status of customary international law. *Id.*

[30] Outer Space Treaty, *supra* note 27, art. I.

[31] *Id.* art. II.

[32] *Id.* art. VI.

B. National Space Mining Laws

On November 25, 2015, US President Barrack Obama signed the Commercial Space Launch Competitiveness Act (CSLCA) into law.[33] The CSLCA addresses aspects of both the private and public space sectors in the US, including commercial launch license streamlining, remote sensing reform, space commerce, space traffic management, the International Space Station and NASA's Space Launch System. The most significant section of the CSLCA for this discussion is Title IV, "Space Resource Exploration and Utilization."[34] Section 51303 of the US Code chapter established by Title IV states:

> "[a] United States citizen engaged in commercial recovery of an asteroid resource or a space resource under this chapter shall be entitled to any asteroid resource or space resource obtained, including to possess, own, transport, use, and sell the asteroid resource or space resource obtained in accordance with applicable law, including the international obligations of the United States."[35]

With respect to the international obligations of the US pursuant to the OST, Congress included a provision confirming that as a result of this legislation, "the United States does not thereby assert sovereignty or sovereign or exclusive rights or jurisdiction over, or the ownership of, any celestial body."[36]

At the time, US space mining companies reacted positively to the Act. Chris Lewicki, President and Chief Engineer of Planetary Resources said,

> [t]his off-planet economy will forever change our lives for the better here on Earth. We celebrate this law as it creates a pro-growth environment for our emerging industry by encouraging

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[33] U.S. Commercial Space Launch Competitiveness Act, Pub. L. No. 114-90, 129 Stat. 70 (2015).

[34] *Id.* § 401-03.

[35] 51 U.S.C. § 51303 (2012).

[36] U.S. Commercial Space Launch Competitiveness Act, § 403.

> private sector investment and ensuring an increasingly stable and predictable regulatory environment.[37]

Moon Express co-founder and executive chairman Naveen Jain said:

> Today's signing is a giant leap for mankind and Moon Express. I am super excited that President Obama has recognized the rights of Moon Express to harvest and own lunar resources that can be used for the benefit of humanity. The Moon is our sister planet that has an incredible amount of natural resources, including helium-3 that can provide clean and limitless fusion energy for the entire world.[38]

Although the Act was widely lauded by US companies, others expressed concern and skepticism. At the April 2018 meeting of the Legal Subcommittee of the United Nations (UN) Committee on the Peaceful Uses of Outer Space (UNCOPUOS) views were expressed by some nations "that [only] the international community of States has jurisdiction over space resources, as well as the right and duty to develop an appropriate international legal framework for such activities."[39] Speaking at the 2016 Galloway Symposium on Critical Issues in Space Law, former US Department of State Legal Adviser Brian Egan spoke about the US interpretation of Article II of the OST. Mr. Egan described how a former Secretary of State noted that while outer space is not subject to national appropriation, that prohibition ends once resources are not "in place." [40] This interpretation permits commercial companies to claim ownership over celestial resources when an extraction has physically taken place, but not at any time beforehand (such as extraction location surveying).

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[37] *President Obama Signs Bill Recognizing Asteroid Resource Property Rights Into Law*, PLANETARY RESOURCES (Nov. 25, 2015), https://www.planetaryresources.com/2015/11/president-obama-signs-bill-recognizing-asteroid-resource-property-rights-into-law/.

[38] *President Obama Signs Law Allowing Moon Express Lunar Mining Rights*, MOON EXPRESS (Nov. 25, 2015), http://www.moonexpress.com/news/president-obama-signs-law-allowing-moon-express-lunar-mining-rights/.

[39] Comm. On the Peaceful uses of Outer Space, Rep. of the Legal Subcomm. on Its Fifty-Seventh Session, ¶ XIII.250, U.N. Doc. A/AC.105/1177 (2018) [hereinafter Legal Subcomm. 2018 Report].

[40] Brian J. Egan, The Next Fifty Years of the Outer Space Treaty, Remarks at the Galloway Symposium on Critical Issues in Space Law (Dec. 7, 2016), *available at* https://2009-2017.state.gov/s/l/releases/remarks/264963.htm.

Or, as George Sowers, a professor at the Colorado School of Mines puts it: "[i]f I'm a US company, the only law I am obligated to follow is US law."[41] In counterpoint, Stephen Hobe, director of the Institute of Space Law at the University of Cologne, argues, "outer space and all non-man-made objects it entails are subject to international regulation, I repeat international regulation, not national regulation."[42]

For better or worse, one other nation has matched the US legislation and, implicitly, the US interpretation of Article II. The European space-focused nation of Luxembourg adopted a law on July 13, 2017 that also permits the extraction, ownership and sale of space resources by private companies.[43]

C. Possible Paths to Governance of Private Space Mining

While it is clear that there is an active debate on the legality and regulation of private space mining, legal solutions are hard to come by. There are essentially two legal models upon which solutions may be built.

i. The Moon Agreement

One path to ensure the international governance of space mining is to reconsider an unpopular international treaty. The Agreement Governing the Activities of States on the Moon and Other Celestial Bodies (Moon Agreement) was finalized in 1979.[44] While the OST has been ratified by 109 nations as of January 1, 2019, the Moon Agreement has only been ratified by 18 as of the same date.[45]

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[41] Debra Werner, *Space Law Workshop Exposes Rift in Legal Community Over National Authority to Sanction Space Mining*, SPACENEWS (Apr. 17, 2018), https://spacenews.com/space-law-workshop-exposes-rift-in-legal-community-over-national-authority-to-sanction-space-mining/.

[42] *Id.*

[43] Loi 674 du 20 juillet 2017 sur l'exploration et l'utilisation des resources de l'espace [Law 674 of July 20, 2017 on the Exploration and Use of Space Resources], JOURNAL OFFICIEL DU GRAND DUCHE DE LUX, July 28, 2017, *available at* http://legilux.public.lu/eli/eta1/leg/loi/2017/07/20/a674/jo
[hereinafter Luxembourg Space Resource Law].

[44] Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, Dec. 18, 1979, 1363 U.N.T.S. 3 [hereinafter Moon Agreement].

[45] Comm. On the Peaceful uses of Outer Space, Rep. of the Legal Subcomm. on the Status of International Agreements Relating to Activities in Outer Space as at 1 January 2019, U.N. Doc. A/AC.105/C.2/2019/CRP.3 (Apr. 1, 2019).

None of the States party to the Moon Agreement have launched astronauts into outer space, though a number of them may be considered spacefaring.[46]

Article 11 of the Moon Agreement states in part:

> [n]either the surface nor the subsurface of the moon, nor any part thereof or natural resources in place, shall become property of any State, international intergovernmental or non-governmental organization, national organization or non-governmental entity or of any natural person.[47]

Article 11 also stipulates that an international regime be established "to govern the exploitation of the natural resources of the moon as such exploitation is about to become feasible."[48] Under the agreement, any resources extracted from the Moon are the common heritage of humankind.[49] The extraction process itself is to be governed pursuant only to an international regime.[50] If either the US or Luxembourg had ratified the Moon Agreement, in order to be valid under international law, their space mining legislation would only be able to permit space resource extraction from asteroids. Pushing for further adoption of the Moon Agreement by nations is one possible way to implement international regulation of commercial space mining activities, but it will surely be an uphill battle to convince States that space mining needs to be governed internationally.

ii. The Building Blocks

Another possible solution is the creation of a new international framework that exclusively governs space mining. In 2016, the Institute of Air and Space Law at the Leiden University in the Netherlands established The Hague International Space Resources Governance Working Group (The Hague Working Group) to begin developing requirements for a framework that could eventually take

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[46] *Id.* Nations that have ratified the Moon Agreement include: Armenia, Australia, Austria, Belgium, Chile, Kazakhstan, Kuwait, Lebanon, Mexico, Morocco, the Netherlands, Pakistan, Peru, Philippines, Saudi Arabia, Turkey, Uruguay and Venezuela.

[47] Moon Agreement, *supra* note 44, art. 11(3).

[48] *Id.* art. 11(5).

[49] *Id.* art. 11(3).

[50] *Id.* art. 11(5)–(8).

the form of a binding treaty or non-binding guidelines. On September 13, 2017, The Hague Working Group published the final product of their first phase of discussions, a document titled "Draft Building Blocks for the Development of an International Framework on Space Resource Activities"[51] (Building Blocks). The first Building Block states that an international space mining framework should "identify and define the relationship of space resource activities with existing international space law, including the provisions of the United Nations treaties on outer space . . . ."[52] The document also states that governmental regulators of commercial space mining "shall adopt a precautionary approach with the aim of avoiding harmful impacts" including with regards to the harmful contamination of celestial bodies, such as the Moon.[53] Most importantly, the building blocks call for an international framework that "create[s] an enabling environment for space resource activities," including by "provide[ing] legal certainty and predictability for operators."[54] However, the work done by The Hague Working Group has not been well received by all. During the April 2018 meeting of the Legal Subcommittee of the UNCOPUOS, the view was expressed that The Hague Working Group's Building Blocks were troubling because:

> fundamental principles of interest to all States had been discussed by a limited group of individuals; the Group had made assumptions about the interpretation of international space treaties; and the output of the Group, namely its study, contained language that was strikingly similar to recent provisions of national laws on space resources, while at the same time lacking the practical considerations contained in the work of the Scientific and Technical Subcommittee (e.g., references to the long-term sustainability of outer space activities).[55]

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[51] THE DRAFT BUILDING BLOCKS FOR THE DEVELOPMENT OF AN INTERNATIONAL FRAMEWORK ON SPACE RESOURCE ACTIVITIES (2017), https://www.universiteitleiden.nl/binaries/content/assets/rechtsgeleerdheid/instituut-voor-publiekrecht/lucht—en-ruimterecht/space-resources/draft-building-blocks.pdf [hereinafter BUILDING BLOCKS].

[52] *Id.* ¶ 1.2(a).

[53] *Id.* ¶ ¶ 9, 9(d).

[54] *Id.* ¶ ¶ 1.1, 4.2(h).

[55] Legal Subcomm. 2018 Report, *supra* note 39, ¶ XIII.251.

The Building Blocks are open for comment and will likely be revised. Unfortunately, the work of The Hague Working Group continues to be viewed with skepticism by many nations.

III. INTERNATIONAL MANAGEMENT OF SPACE RESOURCE EXTRACTION

Even after years of intense debate within the UN and numerous reports and papers being published, the legality of private space mining is still in question. The space mining industry will not prosper if there is a distinct lack of agreement between nations regarding the legality of private mining activities, so perhaps it is time for a new perspective on the issue.

Instead of trying to come to agreement on the legality of private space mining, the international community should begin discussing management solutions. There will always be critics of commercial space mining but discussing and eventually implementing management systems may relieve some concerns. This section will discuss how private space mining could be managed and when the appropriate time to implement a management solution would be.

As mentioned in Section II, Article II of the OST prohibits national appropriation of any celestial body, which includes asteroids.[56] In this way, the law governing outer space is comparable to the law concerning the high seas. Pursuant to international treaty, no nation may appropriate any part of the ocean outside of its territorial sea, which extends no further than twelve nautical miles from its coast.[57] Despite the sovereignty prohibition, Article I of the OST affirms the freedom of all nations to explore and use all areas of outer space.[58] Similarly, freedom to explore the open ocean is also a right of all States, albeit with restrictions from the UN Convention on the Law of the Sea (UNCLOS).[59]

Article 116 of the UNCLOS authorizes States to fish on the high seas in accordance with the provisions of the treaty.[60] Fishing

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[56] Outer Space Treaty, *supra* note 27, art. II.

[57] United Nations Convention on the Law of the Sea art. 3, Dec. 10, 1982, 1833 U.N.T.S. 397 [hereinafter UNCLOS].

[58] Outer Space Treaty, *supra* note 27, art. I.

[59] UNCLOS, *supra* note 57, art. 87.

[60] *Id.* art. 116.

in international waters is managed by Regional Fisheries Management Organizations (RFMOs), which implement catch quotas for different regions of the ocean and for different species of fish. If the international community agrees that all States have the right to extract space resources, a similar management regime could be established. No State should have any reason to object to granting this basic right to all other nations. To ensure that no State overexploits space resources, an international agreement could impose restrictions on the number of resource extractions at one source, either on an asteroid or on the Moon. As mentioned in Section II of this paper, Article I of the OST states in part that "…the use of outer space, including the Moon and other celestial bodies, shall be carried out for the benefit and in the interests of all countries . . . ."[61] Therefore, the exploitation of all the harvestable resources on an asteroid or on regions of the Moon by only a few companies, without any consideration to nations not participating in the resource extractions, would likely violate Article I. Limiting the number of extractions from one site on an asteroid or on the Moon would give other companies and nations equal opportunities to extract resources. Additionally, different minerals could be subject to tighter or looser extraction quotas based on their in-space economic and practical value.

Another management system can be found within the International Telecommunication Union (ITU), a UN agency which regulates satellite orbits and radio frequencies. Part of the mission of the ITU Radiocommunication Sector is "to ensure rational, equitable, efficient and economical use of the radio-frequency spectrum by all radiocommunication services, including those using satellite orbits . . . ."[62] The ITU fulfills its mandate by allocating and assigning radio frequencies to specific space services, such as maritime, mobile, broadcasting and amateur services. This complex process ensures the optimal and fair use of radio frequencies. A similar allocation and assignment system could be used to provide the same protections to space resources. Different minerals could be assigned

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[61] Outer Space Treaty, *supra* note 27, art. I.

[62] *Radiocommunication Mission Statement,* INT'L TELECOMM. UNION, https://www.itu.int/en/ITU-R/information/Pages/mission-statement.aspx (last visited Aug. 10, 2019).

to different uses, and there could be some sort of international registry of different classifications of space resources and limitations on extractions of certain resources. Any legally binding space mining document could establish such a system. Of course, this will likely be nearly impossible to implement and enforce until space mining operations are commonplace.

Besides fisheries and telecommunications, another sector analogous to space mining is deep sea mining. The UNCLOS also established provisions for deep sea mining through the International Seabed Authority.[63] Part of the Authority's mandate is to "promote the orderly, safe and responsible management and development of the resources of the deep seabed area for the benefit of mankind as a whole."[64] The Authority is currently working to complete the "Mining Code," a set of international regulations allowing private companies to extract deep-sea minerals.[65] Furthermore, the Authority has permitted 29 companies to harvest minerals from select areas of the ocean since 2001.[66] However, there are serious environmental concerns being raised by organizations such as Greenpeace with regards to deep sea mining.[67] A similar regime could be established for space mining, whereby private companies, under the supervision of their State government, are granted licenses from an international authority that permits resource extraction from asteroids or the Moon.

Establishing an international body to govern space resources makes sense from a historical standpoint as it would follow in the footsteps of the International Seabed Authority and the ITU, but it

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[63] UNCLOS, *supra* note 57, art. 156.

[64] *Frequently Asked Questions*, INT'L SEABED AUTH., https://www.isa.org.jm/frequently-asked-questions-faqs (last visited Aug. 23, 2019).

[65] *Ongoing Development of Regulations on Exploitation of Mineral Resources in the Area*, INT'L SEABED AUTH., https://www.isa.org.jm/legal-instruments/ongoing-development-regulations-exploitation-mineral-resources-area (last visited Aug. 25, 2019); Legal and Technical Commission, Draft Regulations on Exploitation of Mineral Resources in the Area, International Seabed Authority, ISBA/24/LTC/WP.1/Rev. 1 (July 9, 2018).

[66] *Deep Sea Minerals Contractors,* INT'L SEABED AUTH., https://www.isa.org.jm/deep-seabed-minerals-contractors (last visited Aug. 10, 2019).

[67] Press Release, Greenpeace International, Scientists Sound Alarm about "Destructive" Deep Sea Mining as Greenpeace Demands Government Action (July 23, 2019), *available at* https://www.greenpeace.org/international/press-release/23390/scientists-sound-alarm-about-destructive-deep-sea-mining-as-greenpeace-demands-government-action/.

could also result in an agency that does not have a clear mission, which would result in either overstepping, or undermanaging and thereby harming a nascent industry. The US has not ratified UNCLOS, and therefore is not a Member State of the International Seabed Authority. As such, it is likely the US would be hesitant to ratify an agreement establishing a space mining authority if the authority is organized similar to the seabed authority. Furthermore, since the US is the first nation to pass a space mining bill, other nations interested in space mining activities will look to the US for leadership in the industry. Therefore, a refusal of the US to ratify an international space mining agreement may discourage other nations from ratifying as well, if they trust the reasoning of the US decision. For these reasons, an international space mining authority is not the best option for the management of space resource extraction. That said, a 'code of conduct' similar to the Mining Code is a mechanism that would ensure fair governance of mining activities, without needing to establish a new UN agency. Although commercial deep-sea mining has not been as abundant as predicted, there are solid practices that can be applied to space mining that will enable the longevity of the industry.

Since the enactment of the 2015 US Commercial Space Launch Competitiveness Act recognized the potential for individual ownership of space resources, technology and mission developments have increased as companies attempt to be the first to harvest space resources. Unfortunately, no private company is close to extracting any resources and the first prospecting mission themselves are still years away. This has not stopped the international legal community from engaging in intense debates regarding the legality of space mining and how the industry should be governed. While draft guidelines, studies, reports and active dialogue are positive actions, the implementation of a binding international document governing space mining would be a negative action that could stall any space mining missions before they get off the ground.

The international community should not push for the implementation of such a document for two major reasons. First, an appropriate regime to manage the space mining industry in its current state already exists. The OST establishes environmental protection and mission authorization principles, and the US and Lux-